Systems of Linear Equations

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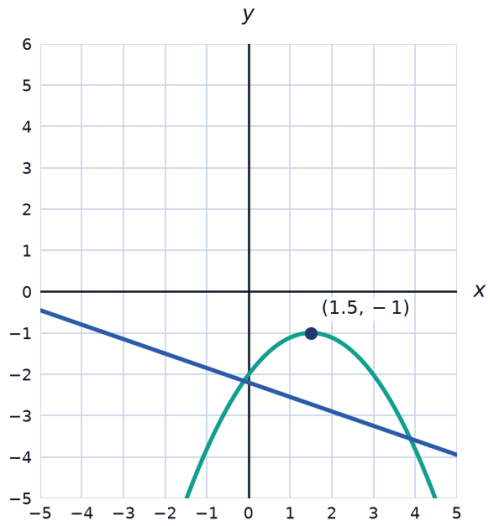
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Setting up the augmented matrix

Write the augmented matrix for the system $2x + y - z = 3$, $x - 3y + 2z = -1$, $4x + y + z = 5$ (give it as its three rows).

2 [Foundational] • Target: Setting up the augmented matrix

Write the augmented matrix for the system $x - y = 4$, $3x + 2y = 7$.



3 [Foundational] • Target: Elementary row operations

List the three types of elementary row operations.

4 [Foundational] • Target: Elementary row operations

Given the augmented matrix with rows $\langle 1, 2, 5\rangle$, $\langle 3, -1, 1\rangle$, apply the row operation $R_2 \to R_2 - 3R_1$ and give the new Row 2.

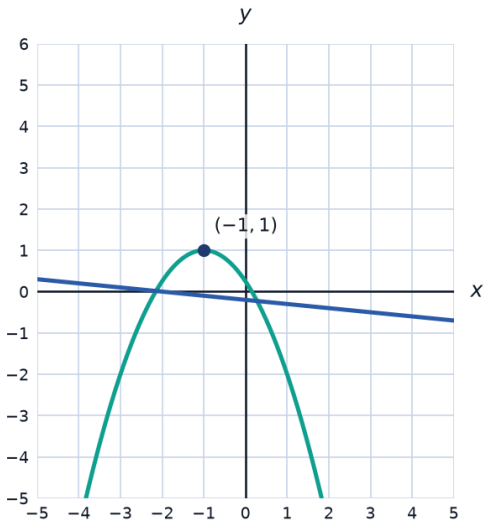
5 [Foundational] • Target: Row echelon form and back-substitution

A system's augmented matrix has been reduced to row echelon form with rows $\langle 1, 2, -1, 4\rangle$, $\langle 0, 1, 3, -2\rangle$, $\langle 0, 0, 1, 5\rangle$ (variables x, y, z). Use back-substitution to solve.

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6 [Foundational] • Target: Row echelon form and back-substitution

Explain why row echelon form makes a system easy to solve, in one or two sentences.



7 [Foundational] • Target: Inconsistent and dependent systems

Solve (or classify) the system $x + y = 3$, $2x + 2y = 8$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inconsistent and dependent systems

For the system $x - 2y = 4$, $-3x + 6y = k$, find the value of k that makes the system have infinitely many solutions (rather than no solution).

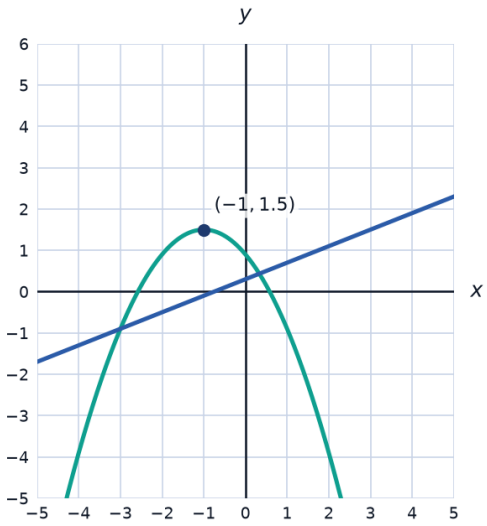
9 [Intermediate] • Target: Why row operations preserve the solution set

Explain briefly why multiplying a row by a nonzero constant c doesn't change a system's solution set.

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10 [Intermediate] • Target: Why row operations preserve the solution set

Why is the restriction $c \neq 0$ needed for this row operation?



11 [Intermediate] • Target: Why row operations preserve the solution set

Explain briefly why adding a multiple of one row to another row also preserves the solution set.

12 [Intermediate] • Target: Systems with a parameter

For what value(s) of k does the system $x + ky = 3$, $2x + 4y = 6$ have infinitely many solutions?

13 [Intermediate] • Target: Systems with a parameter

For what value of k does the system $2x - y = 5$, $4x + ky = 1$ have NO unique solution (i.e., the coefficient matrix is singular)?

14 [Intermediate] • Target: Application: mixture and rate problems

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 L of a 15% solution. Set up the system of equations (let x, y be the liters of each solution used).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

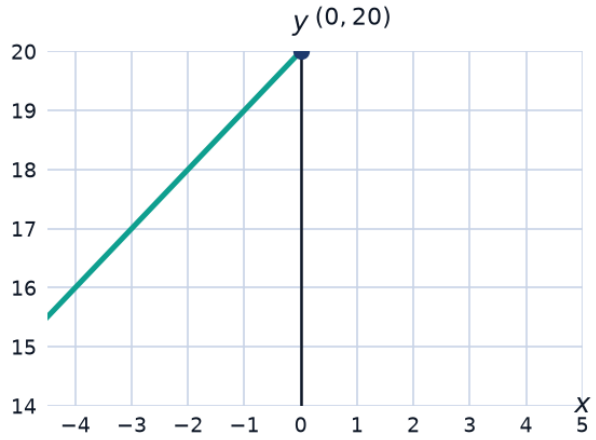
15 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 1.01 Vectors in n Dimensions

Solve the mixture system set up above.

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16 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Two trains leave stations 300 km apart, heading toward each other. One travels at x km/h, the other at y km/h, and they meet after 2 hours: $2x + 2y = 300$. If the faster train's speed is $y = x + 20$, set up and solve the system for x and y.



17 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 4.01 Determinants

A theater sells adult tickets for 12 dollars and child tickets for 7 dollars. On a night with 200 tickets sold for a total of 1900 dollars, set up and solve the system for the number of adult (a) and child (c) tickets sold.

18 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 1.01 Vectors in n Dimensions

Explain why real-world word problems like the two above always translate into a LINEAR system (rather than, say, a quadratic one), given the quantities described.

19 [Challenging] • Target: Setting up the augmented matrix • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Level 1 · FoundationsConvert the matrix into Reduced Row Echelon Form (RREF):

$M = (1,\ -2, 1; 2,\ -4, 5).$

20 [Challenging] • Target: Elementary row operations • Cross-Lesson: 4.01 Determinants

Level 3 · Advanced / University RigorFind the general solution in parametric vector form for the underdetermined linear system: $x_1 + 2x_2 - 3x_3 + x_4 + 2x_5, = 3; -x_1 - 2x_2 + 4x_3 - 2x_4 - x_5, = -2; 2x_1 + 4x_2 - 5x_3 + x_4 + 5x_5, = 7$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Setting up the augmented matrix

Consider the system of linear equations represented by the following augmented matrix: $(1, 2, |, 5; 0, 1, |, 2)$ Use row reduction to determine the unique solution for x and y in the system. What is the value of x?

a. 1

b. 2

c. 3

d. 4

22 [Foundational] • Target: Elementary row operations

Consider the following system of linear equations represented by an augmented matrix: $[1, 2, |, 5; 0, 1, |, 2]$. What is the value of x in the first equation?

a. 1

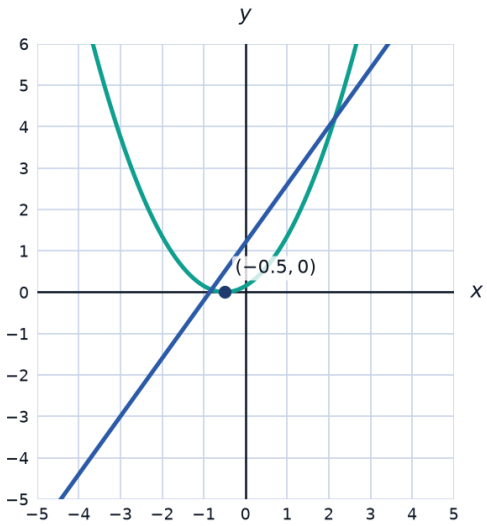
b. 2

c. 3

d. 4

23 [Foundational] • Target: Solving 2x2 systems by elimination

Solve for z: $x + y + z = -1$, $y + z = -4$, $z = 1$ (already row-echelon).



a. 1

b. 3

c. -5

d. 2

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24 [Foundational] • Target: Solving 3x3 systems by Gaussian elimination

Consider the matrix $A = (1, 1, 0; 1, 0, 1; 0, 1, 1)$. Let $\mathbf{b} = (1; 1; 1)$. We want to find the vector $\hat{\mathbf{x}}$ that minimizes $Ax - b$. This is equivalent to finding the projection of $\mathbf{b}$ onto the column space of A, denoted by $\text{Col}(A)$. The normal equations for this least squares problem are $A^T A\mathbf{x} = A^T\mathbf{b}$. The unique solution $\hat{\mathbf{x}}$ to the normal equations yields the best approximate solution. Calculate the value of $\hat{\mathbf{x}}_1$, the first component of $\hat{\mathbf{x}}$.

a. 1/3

b. 1/2

c. 2/3

d. 3/4

25 [Foundational] • Target: Row echelon form and back-substitution

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (2, k\ 1, 3)$ with respect to the standard basis. The image of the vector $\mathbf{v} = (1\ 1)$ under T is $T(\mathbf{v}) = (3\ 4)$. Find the value of the parameter k.

a. 1

b. 2

c. 3

d. 4

26 [Foundational] • Target: Inconsistent and dependent systems

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (1, 2; k, 4)$. The image of the vector $\mathbf{v} = (1; 1)$ under this transformation is $T(\mathbf{v}) = (3; 5)$. Determine the value of the parameter k.

a. 1

b. 2

c. 3

d. 4

27 [Foundational] • Target: Why row operations preserve the solution set

How many solutions does the system $3x + (3)y = 6$, $-6x + (-6)y = -13$ have?

a. no solution

b. exactly one solution

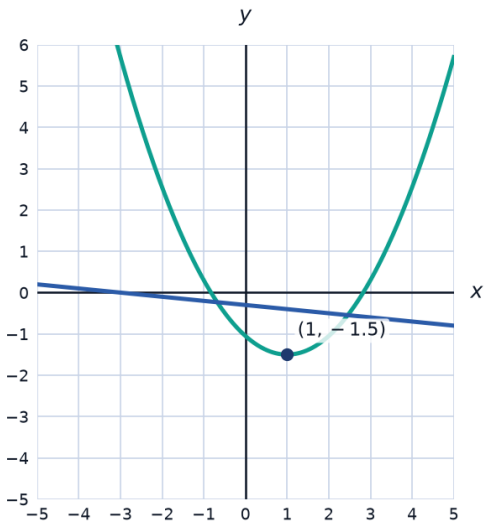
c. infinitely many solutions

d. exactly two solutions

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28 [Foundational] • Target: Systems with a parameter

Consider a matrix $A = (2, -1; 3, k)$. If the determinant of the matrix A^2 is 100, find the positive value of the parameter k.



a. 4

b. 5

c. 2

d. 8

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Application: mixture and rate problems

Consider a linear transformation represented by the matrix $A = (2, 1; 1, -2)$. A point P is mapped to P' by this transformation. If the coordinates of P are $(x; y)$ and the coordinates of P' are $(x'; y')$, then $(x'; y') = A(x; y)$. Suppose a curve is defined by the equation $x'^2 + y'^2 = 5$. What is the equation of the original curve $x^2 + y^2 = k$ in terms of x and y that transforms into this circle? Find the value of k.

a. 1

b. 2

c. 3

d. 4

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30 [Intermediate] • Target: Setting up the augmented matrix

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x, y) = (3x - y, x + 2y)$. Consider the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for $\mathbb{R}^2$, where $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$. Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for $\mathbb{R}^2$, where $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -1)$. If the matrix representation of T with respect to the basis $\mathcal{B}$ is $[T]_{\mathcal{B}} = (a, b\ c, d)$, and $a + d = 5$, determine the value of $\det(T)$.

a. 7

b. 5

c. 3

d. 2

31 [Intermediate] • Target: Elementary row operations

A solid object in $\mathbb{R}^3$ is defined by the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, and the plane passing through the points $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$. If a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ is represented by the matrix $M = (1, 0, 0; 0, 1, 0; 0, 0, 2)$, what is the volume of the transformed solid object?

a. 12

b. 24

c. 36

d. 48

32 [Intermediate] • Target: Solving 2x2 systems by elimination

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (2, 0\ 0, 3)$. What is the image of the vector $\mathbf{v} = (1\ 1)$ under this transformation, i.e., find $T(\mathbf{v})$?

a. (1 1)

b. (2 3)

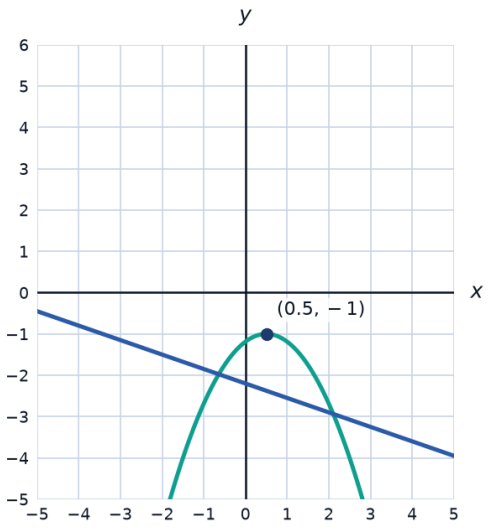
c. (3 2)

d. (4 9)

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33 [Intermediate] • Target: Solving 3x3 systems by Gaussian elimination

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (1, 2; 3, 4)$. If a vector $\mathbf{v} = (2; 1)$ is transformed by T, what is the resulting vector $T(\mathbf{v})$?



a. $(4; 10)$

b. $(2; 3)$

c. $(4; 7)$

d. $(3; 5)$

34 [Intermediate] • Target: Row echelon form and back-substitution

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = (2, 1; 1, 3)$. If $\mathbf{v} = (1; 2)$, find $T(\mathbf{v})$.

a. $(4; 7)$

b. $(3; 7)$

c. $(4; 5)$

d. $(5; 7)$

35 [Intermediate] • Target: Inconsistent and dependent systems

Find the standard matrix of the orthogonal projection onto the line spanned by $\langle 1, 3 \rangle$.

a. $[(1, 3; 3, 9)]$

b. $\left[\left(\frac{9}{10}, -\frac{3}{10}; -\frac{3}{10}, \frac{1}{10}\right)\right]$

c. $\left[\left(-\frac{4}{5}, \frac{3}{5}; \frac{3}{5}, \frac{4}{5}\right)\right]$

d. $\left[\left(\frac{1}{10}, \frac{3}{10}; \frac{3}{10}, \frac{9}{10}\right)\right]$

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36 [Intermediate] • Target: Why row operations preserve the solution set

Which matrix represents the rotation counterclockwise by 60°?

a. $\left[\left(\frac{1}{2}, \frac{\sqrt{3}}{2}; -\frac{\sqrt{3}}{2}, \frac{1}{2}\right)\right]$

b. $\left[\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}; \frac{\sqrt{3}}{2}, \frac{1}{2}\right)\right]$

c. $\left[\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}; -\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)\right]$

d. $[(0, -1; 1, 0)]$

37 [Intermediate] • Target: Systems with a parameter

Which matrix represents the reflection across the x-axis?

a. $[(-1, 0; 0, 1)]$

b. $[(0, -1; 1, 0)]$

c. $[(1, 0; 0, -1)]$

d. $[(1, 0; 0, 1)]$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Application: mixture and rate problems • Cross-Lesson: 1.01 Vectors in n Dimensions

Which matrix represents the reflection across the y-axis?

a. $[(-1, 0; 0, 1)]$

b. $[(1, 0; 0, -1)]$

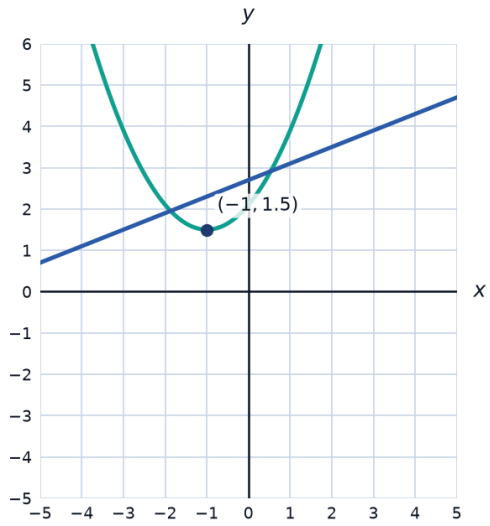
c. $[(0, -1; 1, 0)]$

d. $[(1, 0; 0, 1)]$

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39 [Challenging] • Target: Setting up the augmented matrix • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Which matrix represents the reflection across the line y = x?



a. $[(0, -1; -1, 0)]$

b. $[(0, 1; 1, 0)]$

c. $[(0, -1; 1, 0)]$

d. $[(1, 0; 0, 1)]$

40 [Challenging] • Target: Elementary row operations • Cross-Lesson: 4.01 Determinants

Which matrix represents the orthogonal projection onto the x-axis?

a. $[(1, 0; 0, 0)]$

b. $[(-1, 0; 0, 0)]$

c. $[(0, 0; 0, 1)]$

d. $[(0, -1; 1, 0)]$

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41 [Challenging] • Target: Solving 2x2 systems by elimination • Cross-Lesson: 1.01 Vectors in n Dimensions

Which matrix represents the horizontal shear x ■ x + 3y?

a. $[(1, 0; 3, 1)]$

b. $[(-1, -3; 0, -1)]$

c. $[(1, 3; 0, 1)]$

d. $[(0, -1; 1, 0)]$

42 [Challenging] • Target: Solving 3x3 systems by Gaussian elimination • Cross-Lesson: 3.01 Matrices & Matrix Algebra

S has matrix $[(2, -3; 0, -3)]$ and T has matrix $[(3, -1; -3, -2)]$. Find the matrix of $S \circ T$ (apply T first).

a. $[(15, 4; 9, 6)]$

b. $[(6, -6; -6, 15)]$

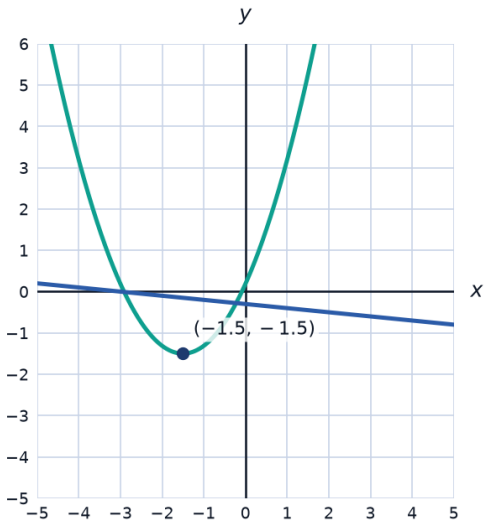
c. $[(5, -4; -3, -5)]$

d. $[(6, 3; 0, 6)]$

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43 [Challenging] • Target: Row echelon form and back-substitution • Cross-Lesson: 4.01 Determinants

Find the standard matrix of $T(x, y) = \langle -2x + 3y, -3x \rangle$.



a. $[(-2, -3; 3, 0)]$

b. $[(-2, 3; -3, 0)]$

c. $[(2, -3; 3, 0)]$

d. $[(-3, 0; -2, 3)]$

44 [Challenging] • Target: Inconsistent and dependent systems • Cross-Lesson: 1.01 Vectors in n Dimensions

For which k are $\langle -2, -4 \rangle$ and $\langle 1, k \rangle$ linearly dependent?

a. 3

b. -2

c. 4

d. 2

45 [Challenging] • Target: Why row operations preserve the solution set • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Let A be a 2×2 matrix given by $A = (2, k; 3, -1)$. If $A^2 = (1, -7; 3, 4)$, find the value of k.

a. 2

b. 3

c. 4

d. 5

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